

Experiment - 1

Comparative Study of Low-Code and Traditional Application Development

Aim:

To study and compare the **Low-Code Development approach** with the **Traditional Application Development approach** based on development time, coding requirements, flexibility, cost, and ease of use.

Definition:

- **Low-Code Development:** A software development approach that uses visual interfaces, drag-and-drop components, pre-built templates, and minimal hand-written code to create applications.
- **Traditional Application Development:** A software development approach in which applications are developed primarily by writing code manually using programming languages and development frameworks.

Comparison:

	Low-Code Development	Traditional Application Development
1.	Uses visual development tools, drag-and-drop components, and pre-built modules.	Uses manual programming and development frameworks.
2.	Requires little or minimal hand-written code.	Requires extensive hand-written code.
3.	Requires basic programming knowledge for many applications.	Requires strong programming and technical knowledge.

4.	UI can be created using predefined components and visual designers.	UI is created manually using programming languages and frameworks.
5.	Applications can be developed quickly.	Development generally takes more time.
6.	Customization may be limited by the capabilities of the platform.	Provides extensive customization and control.
7.	Usually reduces development and maintenance costs.	May have higher development costs due to skilled developers and longer development time.

Result:

The study shows that **Low-Code Development** enables faster application development with less manual coding and is suitable for simple to moderately complex applications. **Traditional Development** requires more time and programming expertise but provides greater flexibility, customization, and control. Therefore, the choice between the two approaches depends on the **complexity, requirements, time, cost, and scalability** of the application